Lab 1: Requirements Elicitation & Modeling

Project: BadNet - Badminton Social Network

1. ASR Documentation

The following table documents the 3 chosen **Architecturally Significant Requirements (ASRs)** and their impact on the system's architectural decisions.

ASR Type	Statement	Architectural Impact
High Scalability (NFR)	The system must handle a surge of concurrent users and match challenges during major tournaments or weekend peaks (expected 2,000+ active users).	Drives the need for efficient database design, caching (ELO ranking, match feed), horizontal scaling, and optimized query performance for real-time challenge and ranking features.
Real-time Processing (NFR)	The system must provide near real-time updates for match results, ELO changes, notifications, and live betting odds.	Requires the use of WebSockets (Laravel Echo + Laravel WebSockets or Pusher), event-driven architecture, and efficient background job processing (Laravel Queues).
Data Integrity & Security (NFR)	The system must securely manage user data, match results, and virtual betting transactions while ensuring accurate ELO calculations and preventing cheating.	Requires strong authentication (Sanctum/Passport), transaction management for ELO & betting, audit logging for match results, and role-based authorization.

2. UML Use Case Diagram